Can AI Counselors Close the Guidance Gap?

Social Image Barriers and the Channel of Educational Support

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Motivation: A Persistent Puzzle

Low-SES students systematically **undermatch**—applying to and attending less selective institutions than their qualifications warrant—even when free guidance is available and the net cost is lower at more selective universities [Hoxby and Turner, 2013]. Majority of high-achieving, low-income (HALI) students apply to zero selective schools [Hoxby and Avery, 2012].

Scale of the problem:

- HALI students apply to selective colleges at **far lower rates** than equally qualified high-income peers; for every 15 High-achieving, high-income (HAHI) applicants there is 1 HAHI applicant; when in fact in the population the ratio is 2.5:1 [Hoxby and Avery, 2013]
- In the UK, students attending one of the four most selective universities earn 30% more than those attending mid-ranking universities by age 30 [Britton et al. 2022].
- Same returns to HALI [Chetty et al. 2020; Van Der Erve, Drayton, and Britton 2021].

Key implication: The barrier is not primarily financial. Something psychological is preventing students from *engaging* with available support.

What Works—and Where It Hits a Ceiling

Light-touch interventions *help* but don't close the gap:

Intervention	Effect	Source
FAFSA simplification	+8pp college completion	Bettinger et al. [2012]
$6 info packet	+42pp application rate	Dynarski et al. [2021]
Peer exposure (UK)	+30% applications to elite	Tadjfar and Vira [2025]
Intensive mentoring	+15-30pp, but expensive	Carrell and Sacerdote [2017]

The residual puzzle: Information matters, but the way it is delivered matters more – because of the presence of behavioral barriers. If barriers seem to be high for HALI students and not financial, barriers could be related to identity/social image.

→ **Hypothesis:** The *channel of delivery* matters. The human counselor in itself is a barrier.

Behavioral Mechanism: Belonging Uncertainty

Core idea: Seeking help from a human counselor is a *socially costly* act for low-SES students.

1. **Belonging uncertainty** [Walton and Cohen, 2011]: Students from underrepresented backgrounds feel the institution is "not for them" $\rightarrow$ they avoid institutional support structures
2. **Social image concerns** [DellaVigna et al., 2012, Bursztyn and Jensen, 2015]: Approaching a counselor means exposing ignorance to an authority figure who can *judge* you
3. **Identity mismatch**: Counselors are typically higher-SES, different ethnicity $\rightarrow$ amplifies the psychological cost of help-seeking

Testable prediction: If we remove the human—and therefore the social evaluation—from the guidance channel, take-up should increase, *especially* among students who face the highest identity costs.

Research Question

Does replacing the human counselor with an AI counselor increase take-up of educational guidance among low-SES students and ultimately lead to more applications to selective schools?

And if so, is this because AI removes:

(a) *all* social evaluation pressure? or

(b) specifically *identity mismatch*?

Experimental Design

Setting: High schools in low-income districts. Decision: university/financial aid applications.

Baseline: *All* students receive an informational SMS (deadlines, resources, reminder that free help exists) except those in T0A.

Randomization (student-level, stratified by school):

Arm	Label	What varies
T0A	No Info	No SMS
T0B	Info Only	Common SMS only (control)
T1	Human Counselor	+ offer of SMS-based human counselor
T2	Matched Human	+ human counselor with shared identity
T3	AI Counselor	+ LLM chatbot, framed as private & judgment-free

Key: The *only* thing that varies across T1–T3 is the *nature of the counselor*. Information, availability, and cost are held constant.

Identification: What Each Comparison Tells Us

- **T0A vs. T0B:** Does offering a information alone increase take-up?
  - If T0B > T0A, the barrier to action is partly explained by information frictions—students don't have full information about the resources available
- **T1 vs. T0B:** Does *any* counseling channel help beyond information alone?
  - If T1 ≈ T0B: barrier is not logistical—students have the info but don't act
- **T3 vs. T1:** Does removing the human increase engagement?
  - If T3 > T1: the barrier is *psychological*, not about access
- **T2 vs. T1:** Does identity matching help?
  - If T2 > T1: part of the barrier is identity mismatch
- **T3 vs. T2:** Is AI better than even a *matched* human?
  - If T3 > T2: the barrier is about *any* human judgment, not just mismatch

Outcomes and Heterogeneity Predictions

Primary outcomes:

1. *Immediate:* Take-up rate (reply to SMS), engagement depth (messages, duration)
2. *Short-term:* Number of university/financial aid applications submitted
3. *Long-term:* University enrollment, selectivity, academic performance

Key heterogeneity prediction (the behavioral signature):

The AI advantage (**T3** − **T1**) should be **largest** for:

- First-generation students
- Ethnic/racial minorities
- Students with low baseline belonging scores

If AI only helps via convenience (24/7), the effect should be *uniform*. Concentration among high-stigma subgroups = evidence for the belonging mechanism.

Simple Model

Student i seeks counseling iff:

$$\underbrace{p_i(1) \cdot V_i}_{\text{expected return}} - \underbrace{c_i}_{\text{logistics}} - \underbrace{\gamma_i \cdot S(\text{channel})}_{\text{social/identity cost}} > p_i(0) \cdot V_i$$

- V_i: value of college admission (lifetime earnings gain)
- $p_i(1), p_i(0)$: probability of admission with and without counseling, $p_i(1) > p_i(0)$
- c_i: logistical cost of seeking counseling (time, effort, scheduling)
- γ_i: sensitivity to social costs (high for low-SES, minority, first-gen)
- $S(\text{channel})$: social/psychological cost of the counseling channel ($S_{\text{human}} > S_{\text{matched}} > S_{\text{AI}} \approx 0$)

Predictions:

- High-SES (γ_i low): T1 $\approx$ T2 $\approx$ T3
- Low-SES (γ_i high): T3 $>$ T2 $>$ T1 $>$ T0
- $\Rightarrow$ **SES gap in take-up** largest in T1, smallest in T3

Power Calculations

Randomization design: schools randomized to Control vs. Treatment; students randomized within.

Comparison	Level	n per arm	What it tests
T0A vs. T0B	Student (within control)	$\sim$380	Does info alone help?
T3 vs. T1	Student (within treatment)	$\sim$380	Does AI beat human?
T2 vs. T1	Student (within treatment)	$\sim$380	Does matching help?
Counseling vs. T0B	School-level	$\sim$1,740	Does any counseling help?

Assumptions: $p_1 = 0.15$, MDE $= 7$pp, $\alpha = 0.05$, power $= 0.80$.
School-level: ICC $= 0.03$, $m = 120$ seniors/school $\Rightarrow$ DEFF $= 1 + 119 \times 0.03 \approx 4.6$.

Required sample:

- **15 treatment schools** ($\sim$1,800 students: 600 per T1/T2/T3)
- **15 control schools** ($\sim$1,800 students: 900 per T0A/T0B)
- **Total: $\sim$30 schools, $\sim$3,600 students**

Contribution & Policy Implications

Contribution to the literature:

- First experimental decomposition of *why* low-SES students avoid guidance: logistical vs. identity vs. general social evaluation barriers
- Bridges social image literature [Bursztyn and Jensen, 2015, DellaVigna et al., 2012] with education access literature [Dynarski et al., 2021, Hoxby and Avery, 2013]
- Low-cost and scalable policy tool (AI counseling) à la [Bergman, 2021].

Policy implications:

- If T3 $>$ T1: AI counseling can unlock human capital that human counselors cannot reach—at near-zero marginal cost
- Especially promising in **developing countries** where human counselors are scarce and social distance between staff and students is large
- Complements (rather than replaces) existing guidance infrastructure

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